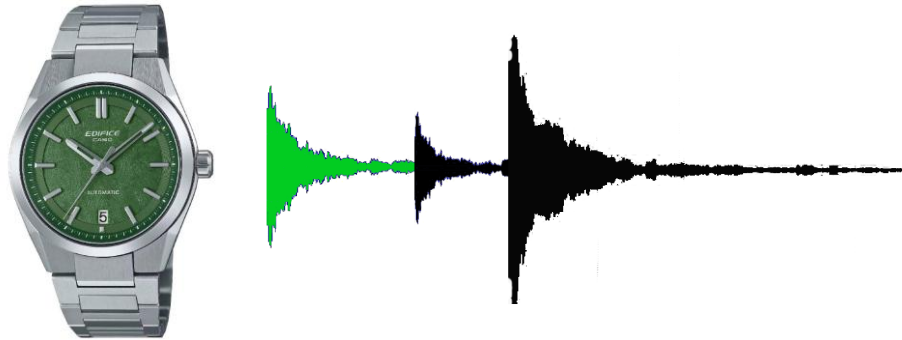




Carnegie Mellon University
Master of
Software Engineering

TimeGrapher Software Architecture

From Tick to Trace: Real-Time Acoustic Analysis



Blue Sky · Team 3 | LG SW Architect Training Program × CMU MSE

Pittsburgh - July 1, 2026

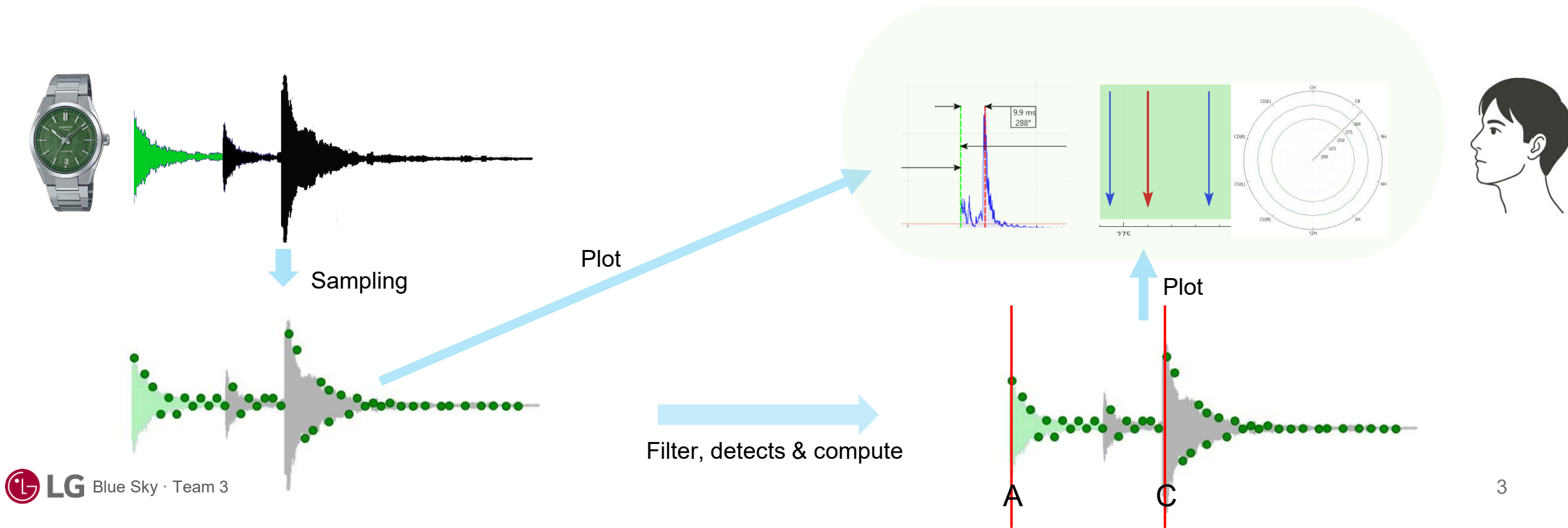
Agenda

- Functional requirements & Constraints
- QA requirements
- Architecture
 - Real-time performance
 - Extensibility
- AI usage
- Lessons learned
- Q&A

Functional requirements & Constraints

- It detects Rate error, Beat error, Amplitude
- Show diagnostics on graphs (> 10 graphs, ...)

Constraints: SW runs on Raspberry Pi 5 Kit (Linux, 16 GB RAM)



QA requirements

Priority	QA	Technical Risk	Experiment
H	Real Time Performance	TR-02, TR-03, TR-04	EXP-01 Dropped Block
H	Low Latency and Low Number of Missed Beats	TR-01, TR-02, TR-03	EXP-02 Segment Timestamp Measurement
M	Extensibility, Modifiability	TR-06, TR-07, TR-08	EXP-03 Tab Extension Cost Measurement
M	Correctness	TR-05, TR-07	EXP-04 Detector Parameter Optimization Under Noise
M	Measurement Accuracy, Error Detection, and Handling	TR-07	EXP-06 Ambient Noise Threshold Validation
L	Long-Term Session Performance	—	EXP-07 Bucket Downsampling Efficiency

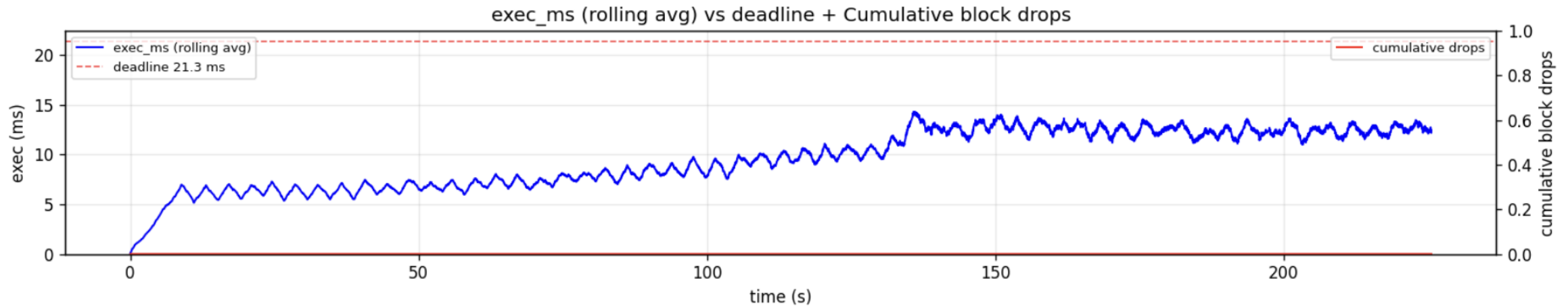
Real-time performance - Scenario

Field	Detail
Source	Mechanical watch placed on microphone
Stimulus	Continuous acoustic beat signal at 28,800 BPH; each 2048-sample audio block arrives from ALSA
Artifact	Audio pipeline — AudioCapture → DSPWorker → MeasurementEngine
Environment	Raspberry Pi 5, 96kHz sample rate (~21 ms block deadline), Qt GUI running concurrently
Response	Every audio block is captured, processed, and a beat measurement emitted without skipping any block
Measure	<i>Deadline miss = 0%; dropped blocks = 0 over a 10-minute session at 96kHz</i>

Real-time performance – Technical Risk

ID	Description	Experiment	ADR	Result
TR-02	43% deadline miss on RPi	EXP-02	ADR-001 DSP Offload Thread 	wait_ms ↓32,000 (420ms → 0.013ms)
TR-03	Signal backlog accumulates	EXP-02	ADR-001 DSP Offload Thread 	Backlog 0% (on RPi)
TR-04	Replot consumes 79% of exec budget	EXP-02	ADR-002 Lazy Rendering 	replot/beat ↓85% (8.22 → 1.20)

Real-time performance – Experiment



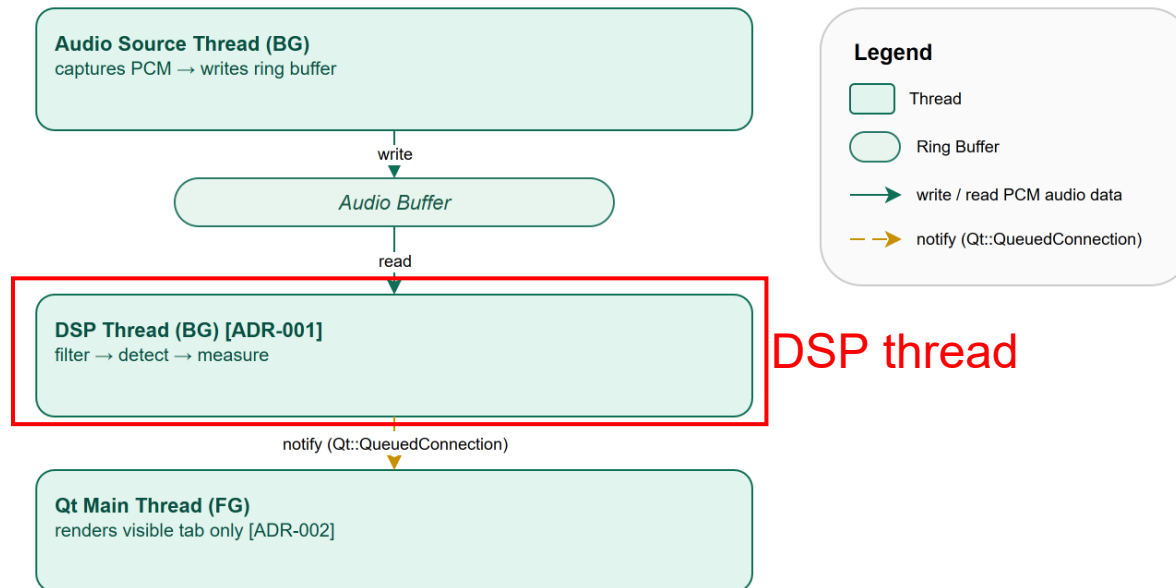
Execution time measurement

Run	sps	exec avg/max (ms)	exec > deadline	Dropped
E2-01	48k	5.8 / 36.6	4.9 %	0
E2-02	96k	9.6 / 39.2	8.1 %	0
E2-03	192k	15.8 / 51.6	12.1 %	0

Real-time performance – Architecture View

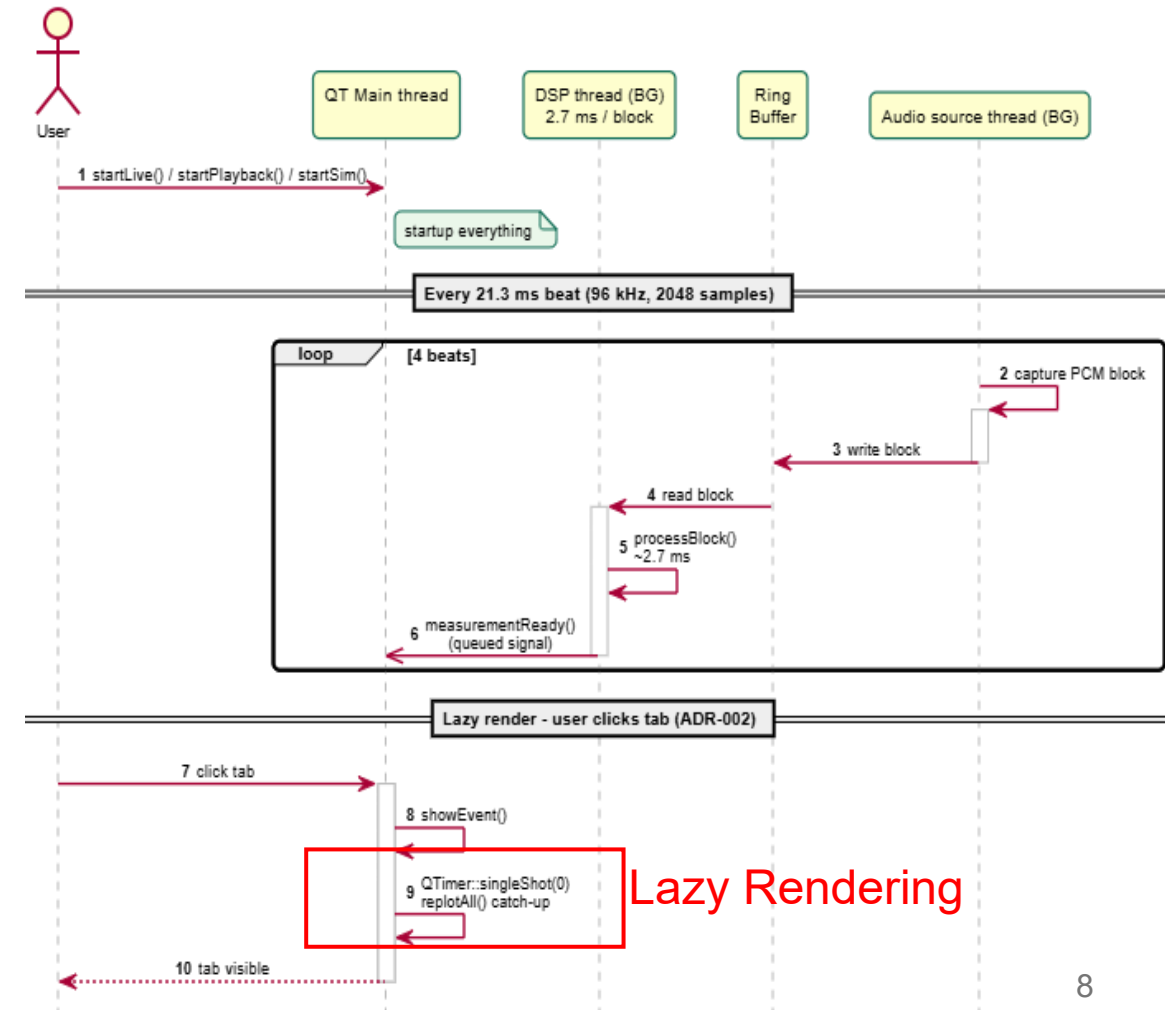
- Introduce new background “DSP Thread”
- GUI thread apply “Lazy rendering”

C&C Thread model view



DSP thread

Sequence Diagram - Lazy Render



Extensibility - Scenario

Field	Detail
Source	Developer (adding a new graph display tab)
Stimulus	Request to implement a new graph type (e.g., Spectrogram, Long-Term Trend, Sequence Display)
Artifact	Source code — Presentation layer and its relationship to layers below
Environment	Development environment (Qt Creator, macOS/Windows), within the 5-week project timeline
Response	New graph widget implemented, registered, and tested independently; no existing modules in lower layers modified
Measure	Files changed outside the new tab itself = ≤ 3 ; references from the new tab to Signal Processing or Acquisition = 0

Extensibility – Technical Risk

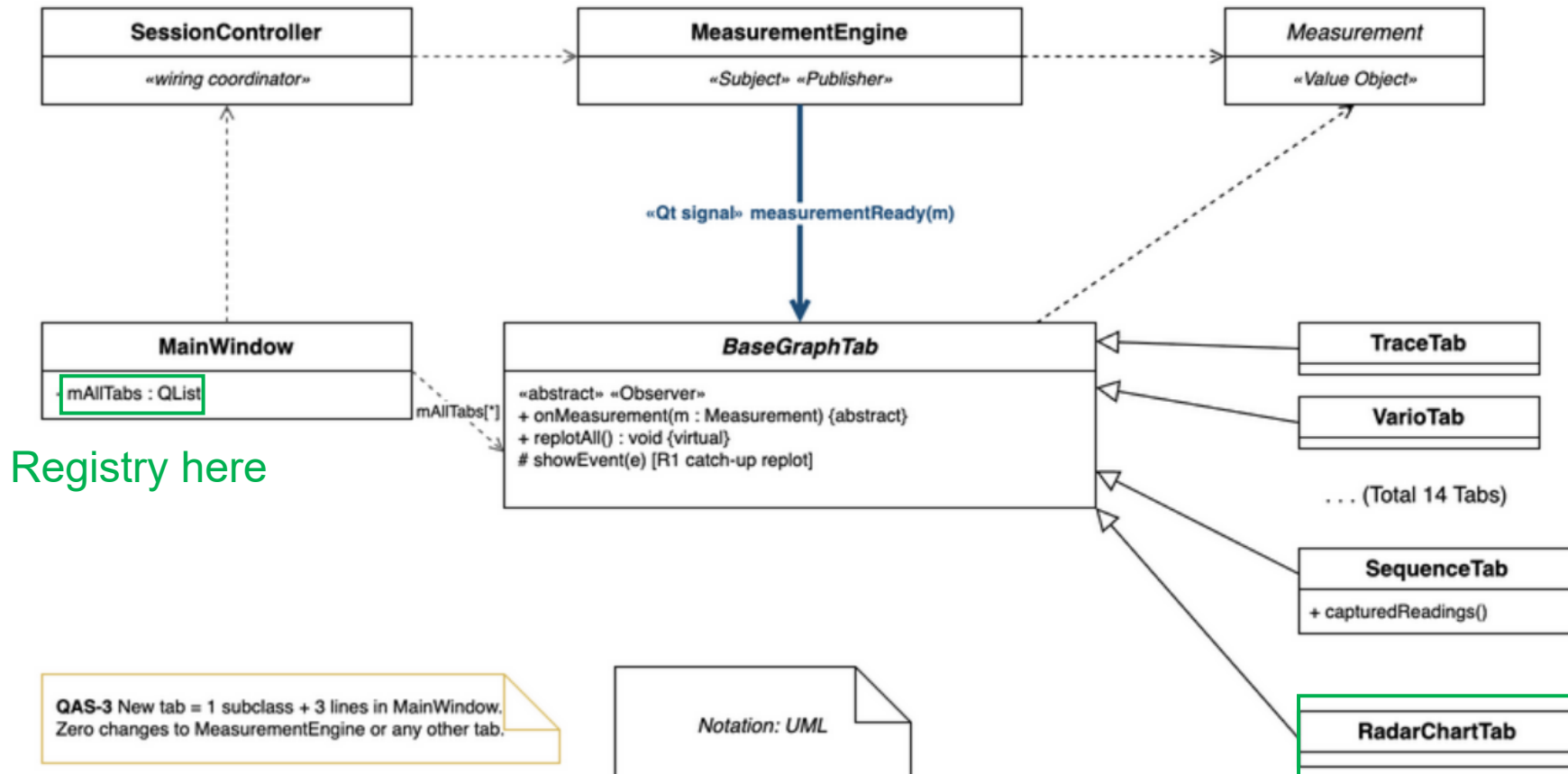
ID	Description	Experiment	ADR	Result
<u>TR-08</u>	Tab addition file-change cost exceeds “ $\leq$ 3-file budget”	<u>EXP-08</u>	BaseGraphTab Observer	<u>ADR-006</u> All 14 tabs added within budget; no lower-layer files touched 

Extensibility – Experiment

Sprint	Tabs added	Trigger	Files changed on a new tab
W2 S1	11 (baseline)	Core requirements	2
W2 S2	13 ++(FilterScopeTab, SweepScopeTab)	Project-plan screen requirements	2
W3 S1	14 ++(RadarChartTab)	Radar/Polar chart (bonus)	3

Extensibility – Architecture View

Graph tab observer contract module view



Registry here

A new tab shall follow contracts of BaseGraphTab

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AI usage – How we use

Agent Skills inputs: *templates, domain knowledge, milestone feedbacks*

Category	AI assistant	Team result
Overall	Understand code base / Requirements quickly	Fast approach to project
Design	Layer decomposition / Module contracts / Surface options & risks	Final architecture decisions
Implementation	Repetitive/formula-based code Witschi math → initial code Experiment log tooling	Review, validate, refine
Documentation	Experiment docs / Design rationale / Architecture notes	Fixed template → AI draft → team edit

AI usage – What AI useful

Category	Task	Useful
Design	Layer decomposition & module contracts	✓ Can help
	Cross-artifact consistency checks	✓ Can help
Coding	Template-conformant code (graph tabs, ADRs)	✓ Can help
Documentation	ADR / experiment / architecture view	⚠ Limited
	Diagram generation	⚠ Limited
Testing	Unit test case generation	✓ Can help

Lessons Learned

✓ What went right	Why
Make “evidence-based” decisions	ADR should be tied to rationales, not opinions
Work in “Parallel” teams	Dev + experiment sub-teams in parallel
Use “AI”	AI assisted analysis scripts, debugging, docs/Team kept quality

✗ What went wrong	Why
Get late requirements review	Demo guidelines & rubric read too late
Work on Dev machine ≠ target	Fine on macOS/Windows; BUT 43% missed deadlines on Rpi

💡 What we would do differently	Why
Start on RPi in week 1	Real-time risk only shows on target hardware; fixes cheaper early
Maintain SW version	Clear traceability for developers and end users

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- **Q&A**



Please let us know your questions...

Thank you!



References

- Milestone 3: <https://github.com/Ji-Min-Lee/2026-3-sw-architect-studio-project/tree/main/docs/milestone3/final>
- Kanban board: <https://github.com/users/Ji-Min-Lee/projects/3/views/2>

Appendix

Extensibility (2)

Adding new features = New files (logic within) + File changes (tab registration)

```
EXPLORER
2026-3-SW-ARCHITECT-STUDIO-PROJECT
src
  tabs
    SweepScopeTab.h
    TraceTab.cpp
    TraceTab.h
    VarioTab.cpp
    VarioTab.h
    WaveformCompTab.cpp Added
    WaveformCompTab.h
  tests
  TimeGrapherTestFilesWeishiMic
  tools
  ui
    DiagnosisDialog.cpp
    DiagnosisDialog.h
    MainWindow.cpp Changed 9+
    MainWindow.h
    MainWindow.ui
    SessionController.cpp
    SessionController.h
    SettingsManager.cpp
    SettingsManager.h
    UserGuideContent.cpp
    UserGuideContent.h
    UserGuideDialog.cpp
    UserGuideDialog.h
    CMakeLists.txt Changed
    Main.cpp

src > ui > MainWindow.cpp > MainWindow(QWidget *)
136 MainWindow::MainWindow(QWidget *parent)
239 // Standard tabs - registerTab() adds a new tab page ANI
242 // SequenceTab owns and wires it, so it is not registered separately.
243 mTraceTab = registerTab(new TraceTab(this), "Trace");
244 mVarioTab = registerTab(new VarioTab(this), "Vario");
245 mSequenceTab = registerTab(new SequenceTab(this), "Sequence");
246 mBeatNoiseScopeTab = registerTab(new BeatNoiseScopeTab(this), "Beat Scope");
247 mBeatErrorTab = registerTab(new BeatErrorTab(this), "Beat Error");
248 mLongTermTab = registerTab(new LongTermTab(this), "Long Term");
249 mEscapementTab = registerTab(new EscapementTab(this), "Escapement");
250 mSpectrogramTab = registerTab(new SpectrogramTab(this), "Spectrogram");
251 mWaveformCompTab = registerTab(new WaveformCompTab(this), "Waveform"); Tab registration
252 mSweepScopeTab = registerTab(new SweepScopeTab(this), "Sweep");
253 mFilterScopeTab = registerTab(new FilterScopeTab(this), "Filters");
254 mRadarChartTab = registerTab(new RadarChartTab(mSequenceTab, this), "Radar");
255 connect(mSequenceTab, &SequenceTab::sequenceUpdated, mRadarChartTab, &RadarChartTab::rebuild);
256
257 // Tab tooltips: show full rubric name on hover
258 auto *tw = ui->GraphicsTabWidget;
259 tw->setTabToolTip(tw->indexOf(ui->RateTab), "Rate / Scope - baseline rate and oscilloscope view");
260 tw->setTabToolTip(tw->indexOf(ui->SoundTab), "Sound Print - acoustic beat waveform image");
261 tw->setTabToolTip(tw->indexOf(mTraceTab), "Trace Display");
262 tw->setTabToolTip(tw->indexOf(mVarioTab), "Rate and Amplitude Stability Over Time");
263 tw->setTabToolTip(tw->indexOf(mSequenceTab), "Multi-Position Sequence Display");
264 tw->setTabToolTip(tw->indexOf(mBeatNoiseScopeTab), "Beat-Noise Scope Display");
265 tw->setTabToolTip(tw->indexOf(mBeatErrorTab), "Beat Error Display and Diagnostic Trace");
266 tw->setTabToolTip(tw->indexOf(mLongTermTab), "Long-Term Performance Graph");
267 tw->setTabToolTip(tw->indexOf(mEscapementTab), "Escapement Analyzer and Marker-Line Display");
268 tw->setTabToolTip(tw->indexOf(mSpectrogramTab), "Time-Frequency Spectrogram Display");
269 tw->setTabToolTip(tw->indexOf(mWaveformCompTab), "Waveform Comparison Display with Timing Markers");
270 tw->setTabToolTip(tw->indexOf(mSweepScopeTab), "Scope Mode with Synchronized Sweep Display");
271 tw->setTabToolTip(tw->indexOf(mFilterScopeTab), "Scope Function with Multiple Filter Views");
272 tw->setTabToolTip(tw->indexOf(mRadarChartTab), "Radar Chart - multi-position watch health overview (bonus)");
```